

Electronics Workshop II, Project-2

Piezoelectric Energy Harvester

Table 5, Nilgiri 104

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Abstract—The project statement is to create a piezoelectric energy harvesting circuit with AC-DC conversion and voltage regulation to power a low-power sensor nodes from vibrations or footsteps. The piezoelectric energy harvesting technique is based on the materials property of generating an electric field when a mechanical force is applied. This phenomenon is known as the direct piezoelectric effect. AC-DC conversion is done by a full-wave rectifier while the voltage regulation is done with the help of a buck converter.

Index Terms—Piezoelectric materials, Buck converter, AC-DC regulator, Low Power devices.

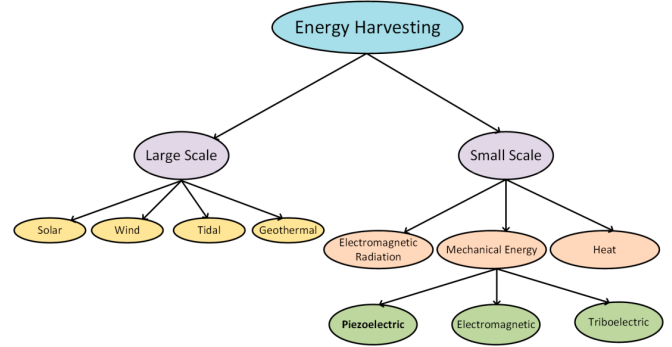


Fig. 1: Classification of energy harvesting sources

I. INTRODUCTION

In recent decades, advancements in wireless technology and microelectronics have enabled the development of wearable devices, including clothing and accessories, powered by batteries or energy harvesting systems. Alongside this, the concept of the Internet of Things (IoT) has emerged, where wireless sensor networks are widely used. IoT has made it possible to deploy smart devices in remote or hard-to-reach locations, such as medical implants inside the human body and infrastructure within smart buildings, where recharging or replacing batteries is challenging.

Although low power integrated circuit technology has improved significantly, the energy density of conventional batteries still falls short of meeting the demands of these applications. As a result, there is a growing need for alternative energy harvesting methods to support self powered systems. Beyond being a practical and cost effective substitute for batteries, energy harvesting also contributes to reducing greenhouse gas emissions and promoting environmental sustainability.

Energy harvesting can be carried out by converting various forms of ambient energy available in the environment, as well as energy generated by humans in their daily activities.

Human body mechanical energy and environmental mechanical energy are widely utilized due to their abundance in everyday life. Any motion in nature can serve as a potential source of kinetic energy. As a result, mechanical energy is one of the most commonly available forms of energy. By harvesting mechanical energy, sufficient power can be generated to support long-term operation of self-powered systems. For instance, approximately 10 mW can be harvested from upper limb movements, about 1 mW from typing activity, nearly 100 μ W from breathing, and up to 1 W can be generated while walking.

We generate up to 1 W. The harvested power density P_{res} for mechanical energy depends on the motion frequency and magnitude, as shown in the resonance power formula:

$$P_{\text{res}} = 4\pi^3 m f_{\text{res}}^3 y Z_{\text{max}} \quad (1)$$

where m is the inertial mass, Z_{max} is the maximum displacement, f_{res} is the resonance frequency, and y is the amplitude of vibration of the housing.

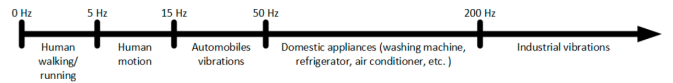


Fig. 2: Frequency level for different mechanical energy sources

II. PIEZO ARRAY

A. Piezoelectric Functionality

The piezoelectric sensor used in this project is typically made of lead zirconate titanate (PZT), which provides higher efficiency compared to natural materials like quartz.

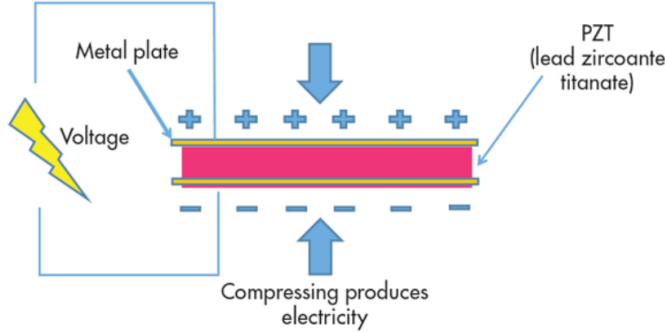


Fig. 3: Working of a piezoelectric sensor

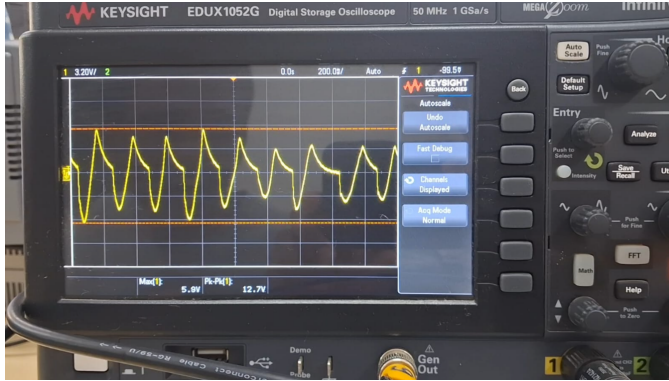


Fig. 4: Output obtained by tapping a single piezo

Piezoelectricity translates to “electricity from pressure.” The prefix piezo in Greek means to press or squeeze. Piezoelectric materials exhibit the piezoelectric effect. This allows a mechanical force to be applied to the material to deform it slightly and the material will produce an output electric power. This is done by rearranging the dipoles of the piezoelectric material. This rearrangement causes a change in the dipole density and therefore a change in the electric field between the dipoles. Because of this change in the electric field, the piezoelectric material produces an output electric power.

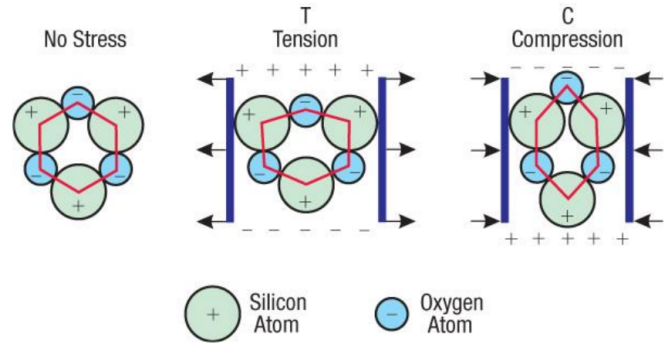


Fig. 5: The piezoelectric effect as observed in Quartz(SiO_2)

B. Piezoelectric Sensor Array Configuration

In order to increase the output voltage and power, multiple piezoelectric sensors were connected in a combination of series and parallel configurations. A total of 20 piezoelectric sensors were used.

Four piezoelectric sensors were connected in series to form a single group. Connecting sensors in series increases the output voltage, as the voltages of individual sensors add up. Five such series groups were then connected in parallel. Parallel connection helps in increasing the output current while maintaining the voltage level.

Thus, the overall configuration consists of 5 parallel branches, each containing 4 piezoelectric sensors in series. This combination ensures that both voltage and current are improved, making the output more suitable for practical applications such as energy harvesting and powering low-power electronic circuits.

The output of a single piezoelectric sensor is typically very low and unstable. Hence, combining multiple sensors helps in improving the overall efficiency and usability of the generated electrical energy.

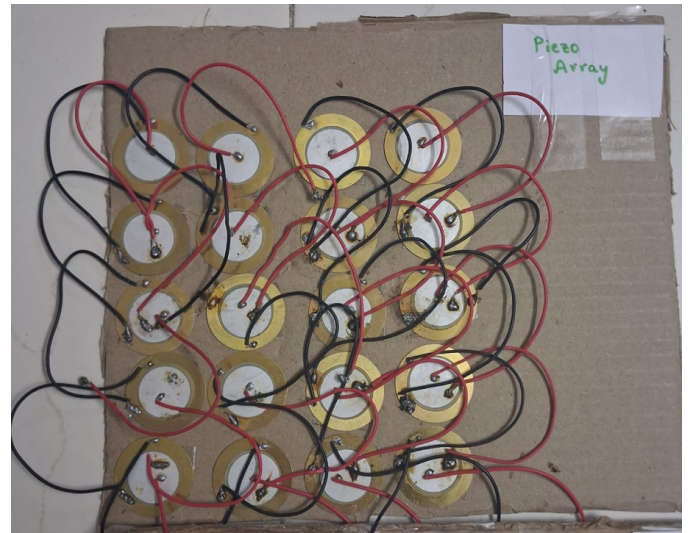


Fig. 6: Piezoelectric sensor setup

C. Demonstration videos

The working of the piezoelectric energy harvesting system is demonstrated through the video titled *piezo_output.mp4*, which can be accessed [here](#).

III. AC-DC CONVERTER

A. Theory of AC-DC converter

The voltage generated by the piezoelectric array is inherently alternating (AC) in nature. To make it suitable for practical applications—especially for powering DC loads—we incorporate an AC–DC conversion stage. This is achieved through rectification using diodes, which allow current to flow only in one direction.

Instead of a half-wave rectifier, we employ a full-wave rectifier. Unlike the half-wave configuration—which utilizes only one half-cycle of the input signal—a full-wave rectifier converts both the positive and negative halves of the AC waveform into a unidirectional output. As a result, it delivers a higher average output voltage, improved power efficiency, and significantly reduced ripple. This makes it more suitable for energy harvesting applications like piezoelectric systems, where maximizing the usable output from a low and variable input is critical.

The resulting output is a pulsating DC signal, which can be further smoothed using filtering components such as capacitors if required.

If we need to rectify AC power to obtain the full use of both half-cycles of the sine wave, a full wave rectifier must be used.

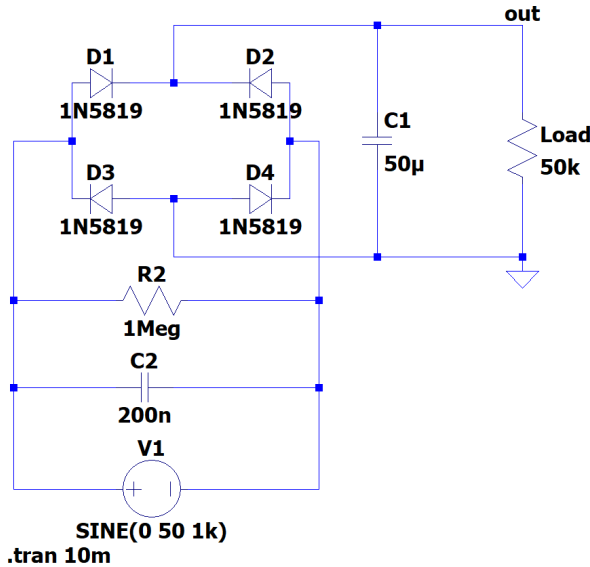


Fig. 7: LTSpice circuit of a full wave rectifier (Resistor and capacitor model the piezo array impedances)

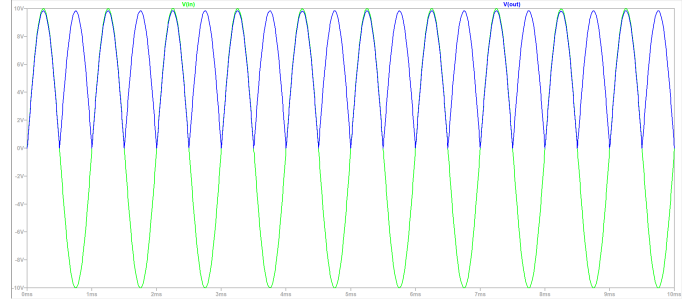


Fig. 8: LTSpice simulation output for full-wave rectifier

The current flow is through two diodes in series for both polarities. Thus, two diode drops of the source voltage are lost ($0.7 \cdot 2 = 1.4 \text{ V}$ for Si) in the diodes. This is a disadvantage which is dealt by using a schottky diode which has a much lower forward voltage drop of 0.2 V to 0.3 V . Therefore the voltage lost comes down to 0.4 V ($0.2 \cdot 2$) to 0.6 V ($0.3 \cdot 2$) which is significantly better than a normal silicon diode.

Current directions for the full-wave bridge rectifier circuit are as shown in the figure below for positive half-cycle and the figure below for negative half-cycles of the AC source waveform. Note that regardless of the polarity of the input, the current flows in the same direction through the load. That is, the negative half-cycle of source is a positive half-cycle at the load.

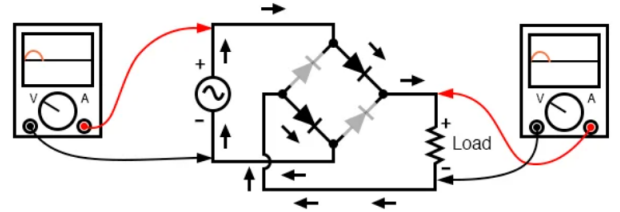


Fig. 9: Full-wave bridge rectifier: Current flow for positive half-cycles

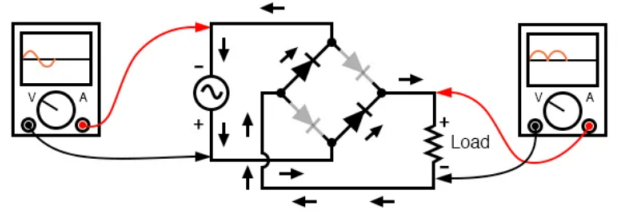


Fig. 10: Full-wave bridge rectifier: Current flow for negative half-cycles

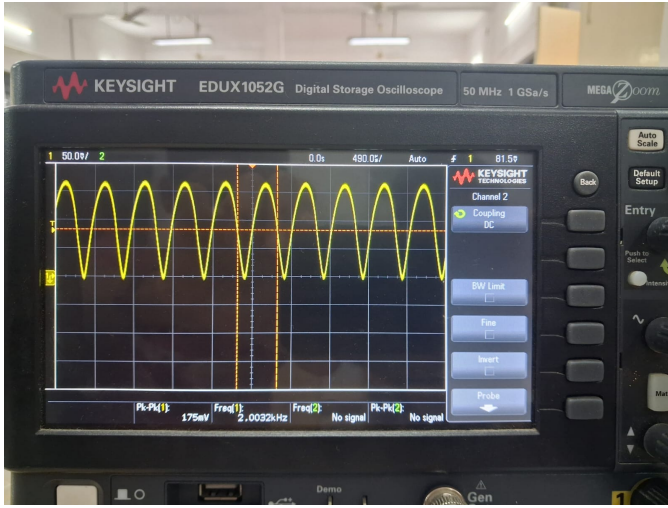


Fig. 11: Full Wave rectification of a sine wave

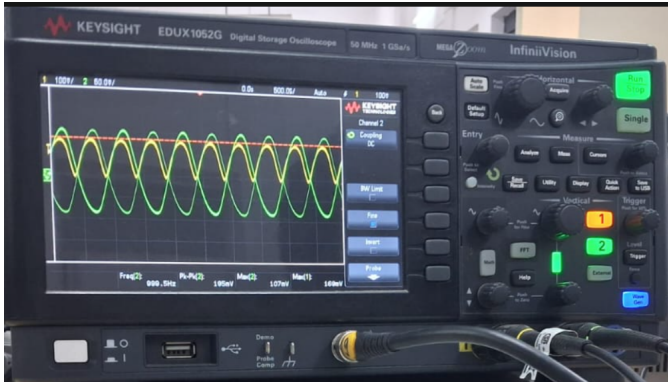


Fig. 12: Full Wave rectification plotted against input

A filter capacitor at the rectifier output smooths the pulsating DC by charging to the peak voltage and discharging slowly between cycles, effectively performing envelope detection to yield a steady DC level.

B. Obtained results

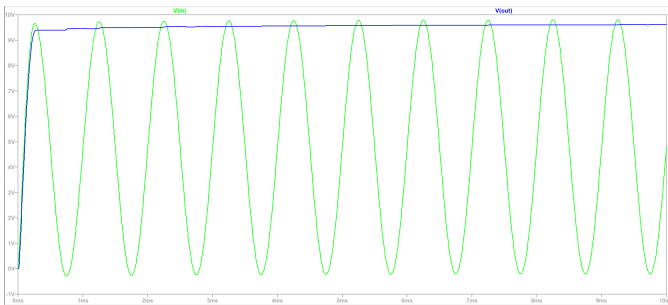


Fig. 13: Full-wave rectified output with filtering in LTspice

From the above simulation result, we can see for 10V AC Amplitude, output is around 9.6V DC. There is only 0.4V drop in the voltage due to 2 Schottky diodes used in series.

C. Demonstration videos

The working of the AC-DC converter is demonstrated through the videos titled *piezo_output_after_AC_to_DC.mp4* and *piezo_output_of_one_series_block_after_AC_to_DC.mp4*, which can be accessed [here](#).

IV. BUFFER

In piezoelectric energy harvesting systems, the source typically exhibits a high output impedance and limited current capability. Directly connecting such a source to a load such as a buck converter will lead to significant loading effects. The switching action of the buck converter draws pulsed current, which can collapse the piezoelectric output voltage, reduce the effective harvested energy, and degrade overall system performance. Therefore, impedance matching is critical.

To address this issue, a buffer stage is introduced between the rectifier output and the buck converter. The buffer is implemented using an LM358 operational amplifier configured as a unity-gain voltage follower. In this configuration, the output is directly fed back to the inverting terminal, while the input is applied to the non-inverting terminal, ensuring that the output voltage closely follows the input voltage.

The key advantage of this configuration lies in its impedance characteristics. The op-amp presents a very high input impedance to the piezoelectric source, thereby minimizing the loading effect and allowing maximum voltage to be extracted. At the same time, it provides a low output impedance to drive the buck converter effectively, ensuring stable operation even under dynamic load conditions. Thus, the buffer stage acts as an isolation interface between the high-impedance energy source and the low-impedance switching load.

The LM358 buffer is powered directly from the piezo rectifier output, not from any external supply. Since the LM358 supports single-supply operation down to approximately 3 V, and the rectified output typically reaches 5–10 V during mechanical excitation, the op-amp is able to self-power from the harvested energy itself. This ensures that the buffer stage remains fully consistent with the self-powered design objective of the system. The LM358 buffer working range of voltage is 3 to 32 V which falls in piezo rectifier output voltage range.

Overall, the buffer stage significantly improves energy transfer efficiency and system stability by decoupling the piezo rectifier stage and the buck controller stage

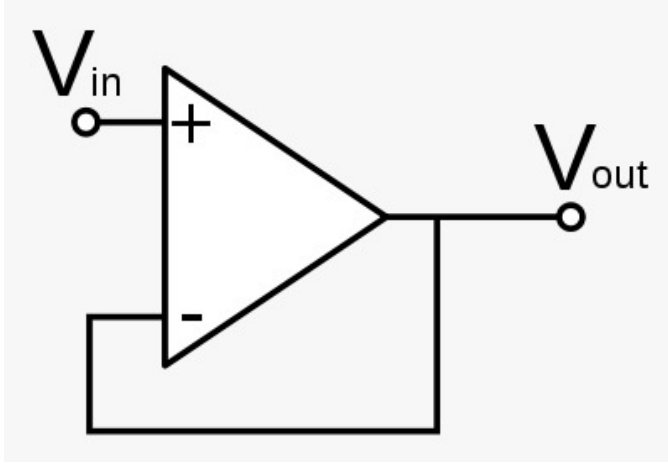


Fig. 14: Circuit Diagram of the Buffer used

V. BUCK CONVERTER

The voltage obtained after rectification is a varying DC voltage. A DC–DC buck converter steps this down to the required constant 3.3 V output using a high-frequency switching power MOSFET (IRLZ44N) controlled by a PWM signal from an Arduino microcontroller.

A. Why Not a Linear Voltage Regulator?

A linear voltage regulator such as the LM7805 could theoretically step down the rectified voltage to a fixed output, but it is highly unsuitable for this application for the following reasons.

- Output voltage **regulated** by **adjusting** the transistor base current
- Power **absorbed** by the load $P_o = V_o I_o$
- Power **dissipated** by the transistor $P_T = V_{CE} I_o$
- Inefficient due to high power dissipation by the transistor.
- Requires a minimum **headroom voltage** $V_{\text{headroom}} = V_{\text{in}} - V_{\text{out}}$ to maintain regulation; for our design $V_{\text{headroom}} = 10 - 3.3 = 6.7$ V, meaning the transistor must withstand and waste an enormous voltage drop, making it completely impractical

To tackle this problem, we model the buck converter as a **switching voltage regulator**, as explained below.

B. Theory of switching voltage regulator

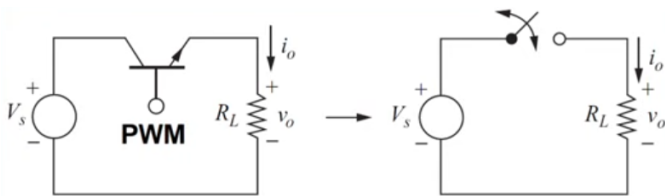


Fig. 15: Buck converter modelled as a switching voltage regulator

The buck converter consists of an N-MOSFET switch, inductor L , capacitor C , freewheeling diode D_1 , and load R . The PWM duty cycle is defined as:

$$D = \frac{t_{\text{on}}}{T} = \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}} \quad (2)$$

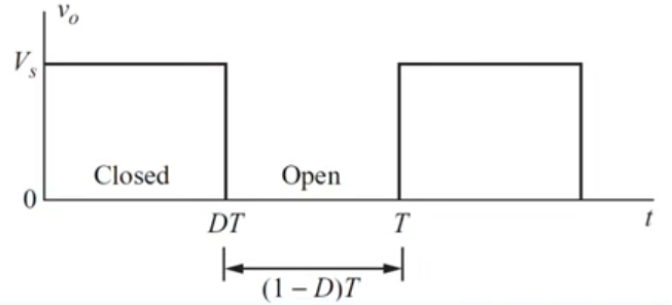


Fig. 16: Duty cycle of a switching voltage regulator

- Gate voltage = 0: switch **open** $\Rightarrow V_{\text{out}} = 0$
- Gate voltage = high: switch **closed** $\Rightarrow V_{\text{out}} = V_{\text{in}}$
- Switch **closed**: diode D_1 is reverse biased
- Switch **open**: diode D_1 forward biased; inductor current freewheels through D_1

C. Derivations and Calculations for Buck converter

It is important to note that the buck converter is designed for a fixed input of $V_{\text{in}} = 10$ V, representing the nominal rectified output under steady mechanical excitation. Under this assumption, a fixed duty cycle of $D = 0.33$ is sufficient to regulate the output to 3.3 V. However, in practice, the piezoelectric source is inherently variable the rectified voltage fluctuates depending on the amplitude and frequency of mechanical input. To account for this, a feedback based bang-bang controller (discussed in next Section) is introduced, which dynamically adjusts the duty cycle in real time to maintain a stable 3.3 V output across varying input conditions.

Output Voltage: The average output voltage is obtained by integrating over one switching period:

$$\begin{aligned} V_{\text{out}} &= \frac{1}{T} \int_0^T V_{\text{out}}(t) dt \\ &= \frac{1}{T} \left[\int_0^{DT} V_{\text{in}} dt + \int_{DT}^T 0 dt \right] \\ &= \frac{V_{\text{in}} \cdot DT}{T} \end{aligned} \quad (3)$$

$$V_{\text{out}} = D \cdot V_{\text{in}} \quad (4)$$

The required duty cycle is therefore:

$$D = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{3.3}{10} = 0.33 \quad (5)$$

Steady-State Inductor Ripple: Since the gate signal is periodic, $i_L(t + T) = i_L(t)$. Volt-second balance requires the average inductor voltage to be zero:

$$V_{L,avg} = \frac{1}{T} \int_t^{t+T} V_L(\tau) d\tau = 0 \quad (6)$$

Switch closed ($0 \leq t \leq DT$):

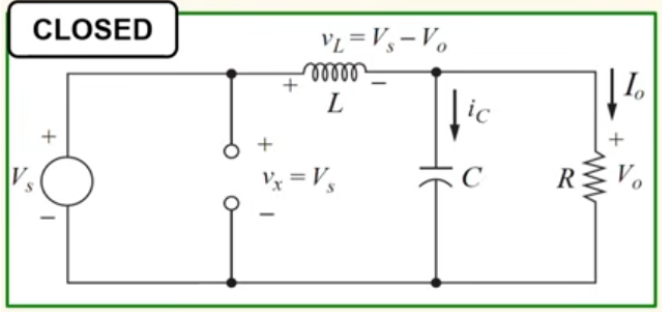


Fig. 17: Buck converter circuit during switch ON (closed) condition

$$V_L = V_S - V_o = L \frac{di_L}{dt} \quad (7)$$

$$\Delta i_{L,closed} = \frac{(V_S - V_o) DT}{L} \quad (8)$$

Switch open ($DT \leq t \leq T$):

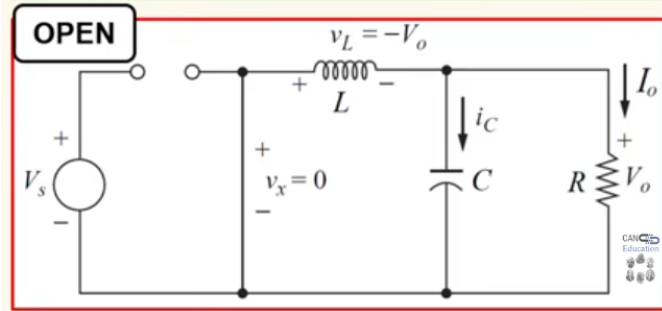


Fig. 18: Buck converter circuit during switch OFF (open) condition

$$V_L = -V_o = L \frac{di_L}{dt} \quad (9)$$

$$\Delta i_{L,open} = \frac{-V_o(1-D)T}{L} \quad (10)$$

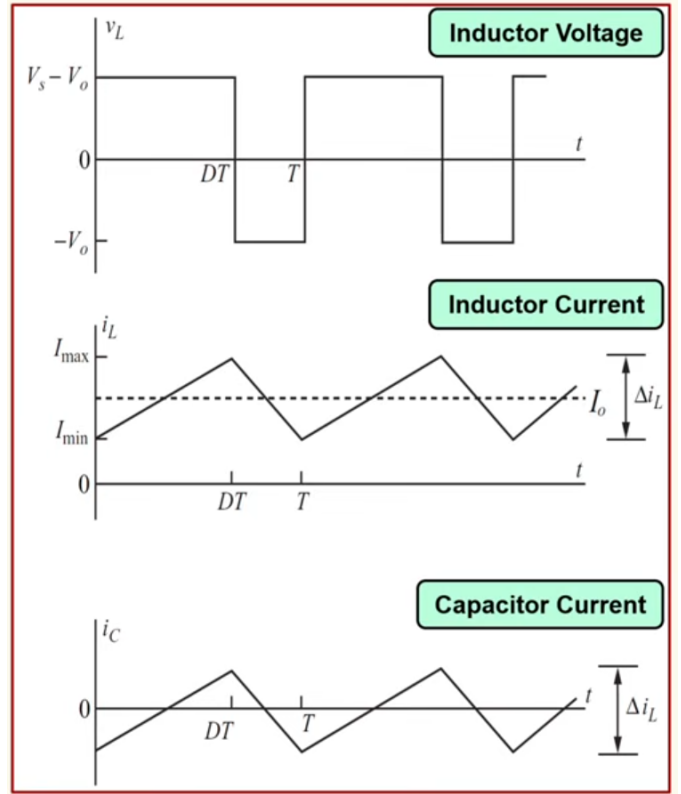


Fig. 19: Buck converter waveforms

Applying the steady-state condition $\Delta i_{L,closed} + \Delta i_{L,open} = 0$ confirms $V_o = DV_S$. The peak-to-peak inductor current ripple is:

$$\Delta i_L = \frac{V_o(1-D)}{L f_s} \quad (11)$$

Inductor Design: At steady state $I_L = I_o = V_o/R$, giving peak and minimum currents:

$$I_{max} = \frac{V_o}{R} + \frac{V_o(1-D)}{2L f_s} \quad (12)$$

$$I_{min} = \frac{V_o}{R} - \frac{V_o(1-D)}{2L f_s} \quad (13)$$

For Continuous Conduction Mode (CCM), $I_{min} > 0$, which gives:

$$I_{min} = \frac{(1-D)R}{2f_s} \quad (14)$$

Taking $\Delta i_L = 5\%$ of $I_o = 1 \text{ A} = 50 \text{ mA}$ and $f_s = 50 \text{ kHz}$:

$$L = \frac{V_o(1-D)}{\Delta i_L \cdot f_s} = \frac{3.3 \times 0.67}{0.05 \times 50 \times 10^3} \approx 884.4 \mu\text{H} \quad (15)$$

We select $L = 890 \mu\text{H}$ (four $220 \mu\text{H}$ + one $10 \mu\text{H}$ inductors in series). The revised ripple becomes:

$$\Delta i_{L,new} = \frac{V_o(1-D)}{L f_s} = \frac{3.3 \times 0.67}{890 \times 10^{-6} \times 50 \times 10^3} \approx 49.6 \text{ mA} \quad (16)$$

The Arduino can provide PWM at up to 62.5 kHz, so $f_s = 50$ kHz is well within range. The inductor RMS current is:

$$I_{L(\text{rms})} = \sqrt{I_L^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}}\right)^2} \approx 2.0001 \text{ A} \quad (17)$$

Capacitor Design: The charge variation equals the area under the i_C vs. t graph:

$$\Delta Q = \frac{\Delta i_L}{8 f_s} \quad (18)$$

Output Voltage Ripple

$$\text{Capacitor current: } i_C = i_L - i_o \quad (19)$$

$$\text{Capacitor charge: } Q = CV_o \quad (20)$$

Change in capacitor charge:

$$\Delta Q = C \Delta V_o \quad (21)$$

$$\Delta Q = \frac{1}{2} \cdot \frac{T}{2} \cdot \frac{\Delta i_L}{2} = \frac{\Delta i_L T}{8} = \frac{\Delta i_L}{8 f_s} \quad (22)$$

Change in output voltage:

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{\Delta i_L}{8 C f_s} \quad (23)$$

$$\Delta i_L = \frac{V_o(1-D)}{L f_s} \quad (24)$$

$$\Delta V_o = \frac{1}{8 C f_s} \cdot \frac{V_o(1-D)}{L f_s} \quad (25)$$

$$= \frac{V_o(1-D)}{8 L C f_s^2} \quad (26)$$

Rearranging with $\Delta V_o/V_o = 1\%$ (target):

$$C = \frac{1-D}{8 L \left(\frac{\Delta V_o}{V_o}\right) f_s^2} \approx 3.76 \mu\text{F} \quad (27)$$

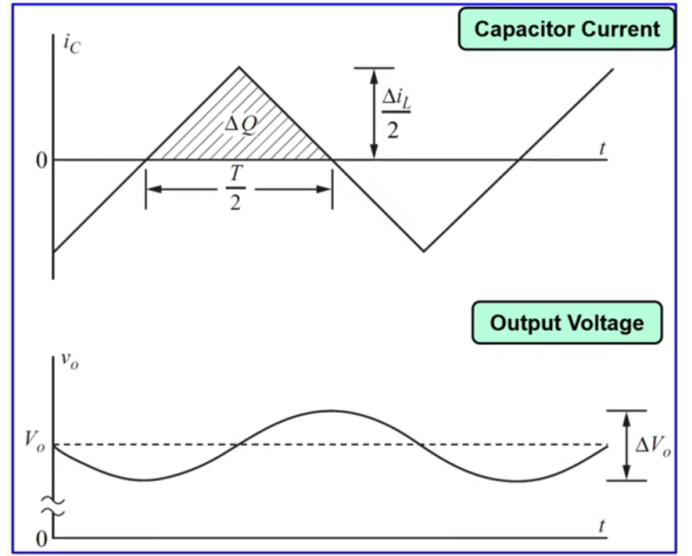


Fig. 20: Output voltage ripple

Capacitor ESR: The maximum allowable equivalent series resistance r_C is:

$$\Delta V_o = r_C \cdot \Delta i_C = r_C \cdot \Delta i_L \quad (28)$$

$$r_C = \frac{\Delta V_o}{\Delta i_L} = \frac{0.033}{0.05} \approx 0.06 \Omega \quad (29)$$

The capacitor RMS current is:

$$I_{C(\text{rms})} = \sqrt{I_{C,dc}^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}}\right)^2} \approx 0.02 \text{ A} \quad (30)$$

TABLE I: Buck Converter Design Values

Parameter	Calculated	Selected
Duty cycle D	0.33	0.33
f_s	50 kHz	50 kHz
L	884 μH (minimum)	890 μH
C	3.76 μF (minimum)	5 μF
ESR r_C	14.3 Ω	—

D. LTSpice Plots

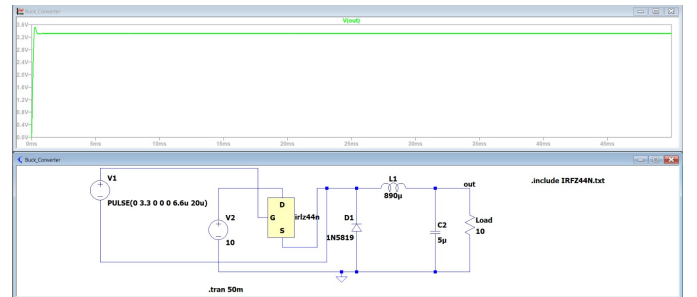


Fig. 21: LTSpice circuit diagram and Output Plot

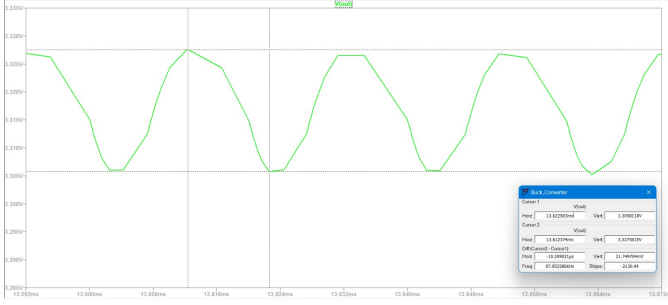


Fig. 22: Output Voltage Ripple

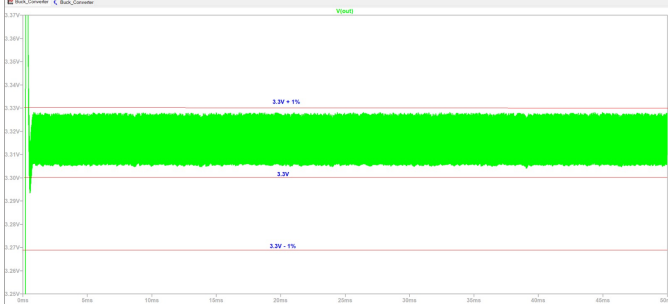


Fig. 23: Plot depicting less than 1% error in V_{out}

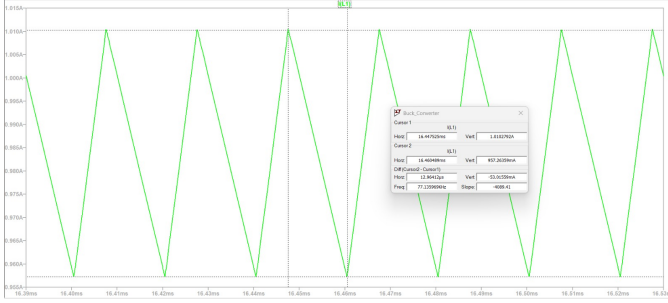


Fig. 24: Inductor Current ripple

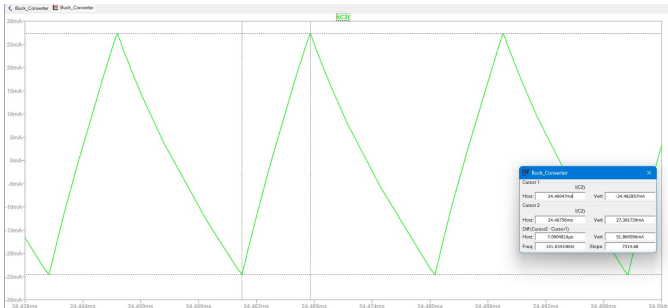


Fig. 25: Capacitor Current ripple

E. Theoretical vs simulation vs hardware

LTspice simulation confirms $\Delta V_o \approx 22 \text{ mV} < 33 \text{ mV}$ (target) and $\Delta i_L \approx 53 \text{ mA} \approx 50 \text{ mA}$ (target), verifying the design meets all the targeted specifications.

TABLE II: Comparison of Theoretical, Simulation, and Hardware Results

Parameter	Theoretical	Simulation	Hardware
Output Current ripple(mA)	50	53	—
Output Voltage ripple(mV)	30	22	—

F. Demonstration videos

The working of the Buck converter is demonstrated through the videos titled *buck_working_with_no_controller_for_20V.mp4*, which can be accessed [here](#).

VI. BUCK CONTROLLER

To regulate the variable DC output obtained from the piezoelectric energy harvesting stage, a buck converter is employed to step down the voltage to a stable 3.3 V. The input to the converter can vary over a wide range (approximately 0–10 V) depending on the mechanical excitation of the piezoelectric source. So, the duty cycle of the PWM needs to be varied accordingly based on this input DC voltage. To drive this PWM duty cycle, a feedback based controller is made using Arduino.

Since the Arduino analog input pins are limited to reading a maximum of 5 V, a resistive voltage divider (6.8 k Ω and 10 k Ω) is used to scale the output voltage to a safe measurable range. This gives a safe measurable output range of 0 to 8.4 V, which is more than sufficient for the target output of 3.3 V. The scaled voltage is sampled using the ADC and then converted back to the actual output voltage using an appropriate scaling factor.

The voltage divider is implemented using a 6.8 k Ω resistor (R_1) in series with a 10 k Ω resistor (R_2), with the ADC input connected across R_2 . The division ratio is given by:

$$V_{ADC} = V_{out} \times \frac{R_2}{R_1 + R_2} = V_{out} \times \frac{10}{6.8 + 10} = V_{out} \times 0.595 \quad (31)$$

The Arduino ADC resolves this scaled voltage over 1023 steps with a 5 V reference, yielding a step size of approximately 4.89 mV per count. The actual output voltage is recovered in firmware by applying the inverse of the divider ratio:

$$V_{out} = \frac{ADC_{raw}}{1023} \times 5.0 \times \frac{1}{0.595} \quad (32)$$

To further improve measurement accuracy, 16 consecutive ADC samples are averaged before applying the scaling factor, effectively suppressing high-frequency noise on the feedback line.

The control strategy employed is a *discrete-time hysteretic (bang–bang) feedback controller with incremental duty-cycle adjustment*. Unlike classical proportional (P), proportional-derivative (PD), or proportional-integral-derivative (PID) controllers, this approach does not compute a continuous error signal or its derivatives. Instead, it operates based on threshold

comparison. It is a simple, two-state (on/off) strategy ideal for high-speed, low-accuracy applications.

Intuitively, the controller behaves like a simple decision system: it continuously checks whether the output voltage is too high, too low, or within an acceptable range. If the voltage is higher than the desired value, the controller reduces the duty cycle of the PWM signal, thereby lowering the output. If the voltage is lower than the desired value, the duty cycle is increased to raise the output. When the voltage lies within a small tolerance band around the target value, no adjustment is made. This creates a stable operating region and avoids unnecessary switching.

Mathematically, the control action can be described as:

if $V_{out} > V_{target} + \Delta$, duty $\downarrow$

if $V_{out} < V_{target} - \Delta$, duty $\uparrow$

otherwise, duty remains unchanged

where $V_{target} = 3.3 \text{ V}$ and $\Delta = 0.03 \text{ V}$.

Additionally, a minimum voltage threshold of 1 V is imposed. If the sensed voltage falls below this level, the controller reduces the duty cycle to a low value, preventing operation under insufficient input power conditions.

The PWM signal controlling the buck converter is generated using Timer1 of the Arduino in fast PWM mode. The duty cycle is updated in discrete steps, enabling real-time feedback control with minimal computational complexity.

This hysteretic control approach is particularly well-suited for piezoelectric energy harvesting systems, where the input is inherently variable. Its simplicity, robustness, and low computational overhead make it an effective solution for maintaining a stable output voltage without the need for complex tuning required in PID-based controllers.

The regulated DC output is maintained at approximately 3.3V with a maximum deviation of $\pm 1\%$, demonstrating satisfactory performance under the given operating conditions.

A. Use of Arduino for Control Implementation

It is important to clarify that the use of the Arduino in this project is primarily for prototyping, validation, and demonstration purposes, rather than as a final deployment solution. While the Arduino requires an external power supply and has non-negligible power consumption, its inclusion allows rapid implementation and testing of the control strategy.

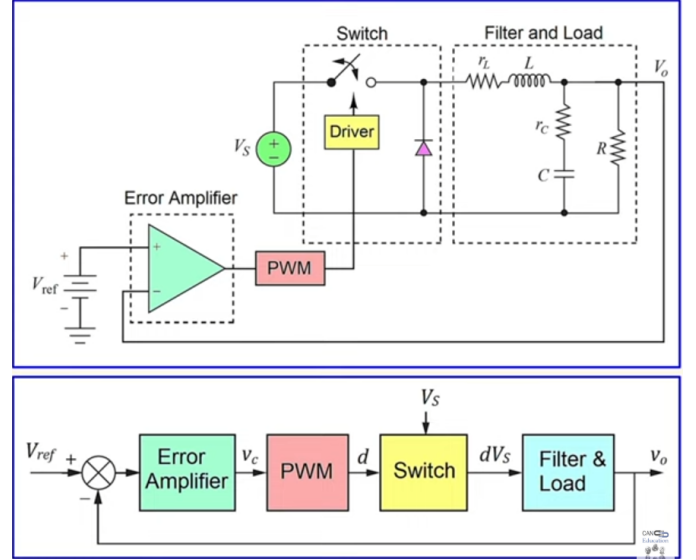


Fig. 26: Block Diagram of a practical Controller Circuit

The objective of this work is to validate the feasibility of regulating a highly variable piezoelectric output using a feedback-controlled buck converter. The Arduino serves as a flexible platform to implement, modify, and observe the behavior of the control algorithm in real time, thereby enabling efficient verification of the design approach.

In practical energy harvesting systems, dedicated buck converter and controller ICs are typically used instead of microcontrollers. These ICs are specifically designed for low-power operation and do not require a continuous external supply. They are capable of cold-start operation, wherein the initial energy harvested from the piezoelectric source is sufficient to power the control circuitry, after which the system sustains itself through the feedback loop.

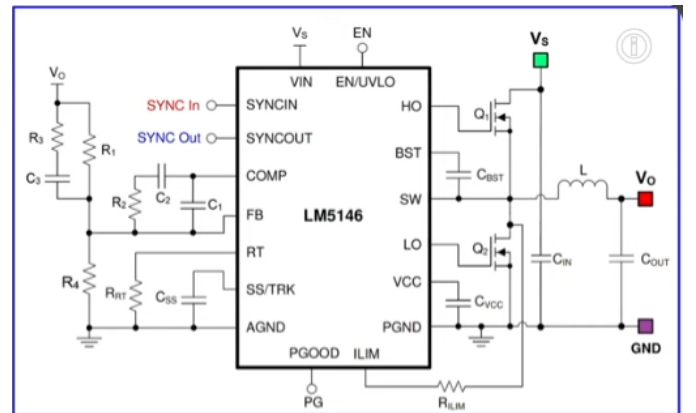


Fig. 27: LM5146 IC Circuit Diagram

Such ICs (like LM5146) incorporate internal control mechanisms, including ramp generators and compensation networks, to regulate the output voltage efficiently. Depending on the design requirements, higher-order control strategies (such as

proportional or PID-based compensation) can be implemented using external components integrated with the IC.

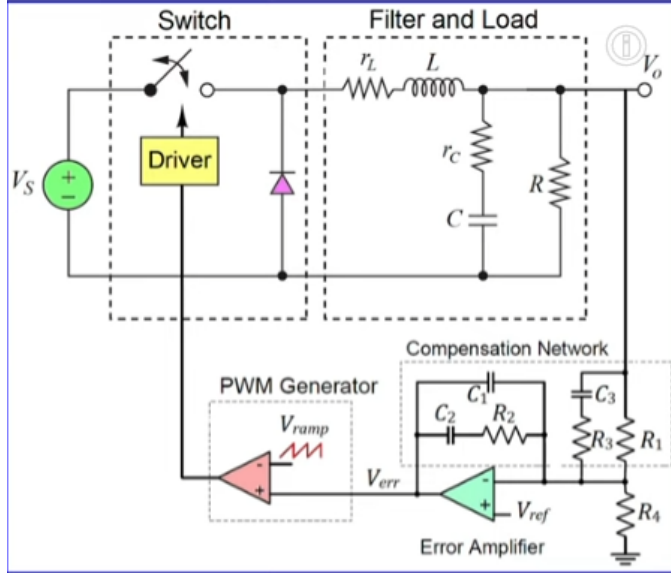


Fig. 28: Example of an higher order controller design

Commercially available evaluation boards, while performance-optimized, are cost-prohibitive (5,000–10,000 rupees). A custom PCB with a dedicated controller IC would offer a compact, cost-effective alternative, but PCB design and fabrication were beyond the scope of this project.

Therefore, the Arduino-based implementation should be viewed as a proof-of-concept platform that demonstrates the validity of the control strategy. The insights gained from this implementation can be directly translated into a low-power, fully integrated hardware solution for real-world piezoelectric energy harvesting applications.

B. Theoretical vs Hardware

Finally, the theoretical analysis predicts that the output voltage settles at 3.3 V. In simulation, the voltage stabilizes at approximately 3.32 V, as demonstrated in the video below which is within the targeted $\pm 1\%$ threshold.

C. Demonstration videos

The working of the Buck Converter with Controller is demonstrated through the videos titled *buck_working.mp4* and *buck_working_with_piezo.mp4* which can be accessed [here](#).

VII. COMPARATOR

A. Op-Amp based comparators

The op-amp comparator is a fundamental analog circuit used to compare two voltage levels. It compares an input voltage with a reference voltage, V_{REF} , and produces a binary output based on the result of this comparison.

In its simplest form, the comparator operates in open-loop configuration, without any feedback. Due to the very high open-loop gain of the op-amp, even a small difference between

the input terminals drives the output into saturation. If the voltage at the non-inverting terminal (V_+) is greater than the voltage at the inverting terminal (V_-), the output saturates to the positive supply rail. Conversely, if $V_+ < V_-$, the output saturates to the ground rail. Mathematically, this can be expressed as:

$$V_{out} = \begin{cases} +V_{CC}, & \text{if } V_+ > V_- \\ 0, & \text{if } V_+ < V_- \end{cases}$$

Depending on how the input and reference voltages are applied, two configurations are commonly used. In a non-inverting comparator, the input signal is applied to the non-inverting terminal and the reference voltage is applied to the inverting terminal, resulting in a high output when the input exceeds the reference. In contrast, in an inverting comparator, the input signal is applied to the inverting terminal and the reference voltage to the non-inverting terminal, producing a high output when the input falls below the reference.

Thus, an op-amp comparator effectively converts an analog input signal into a digital output, making it a key building block in threshold detection and window comparator circuits.

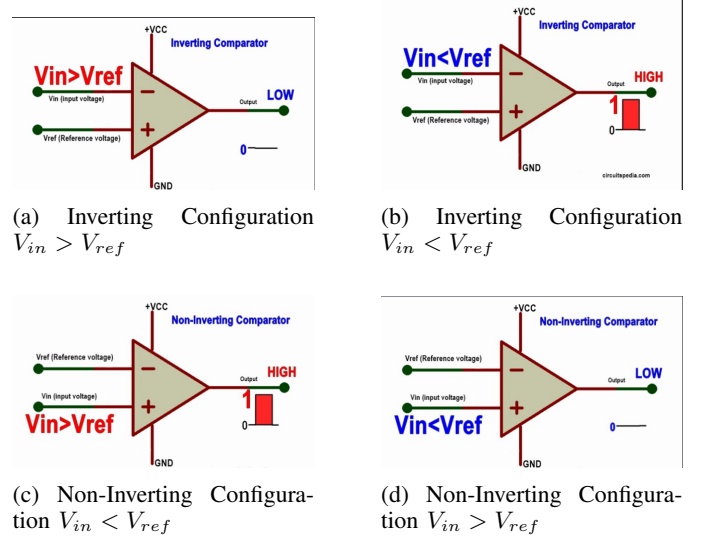


Fig. 29: Op-Amp Comparator topologies

B. Window comparator

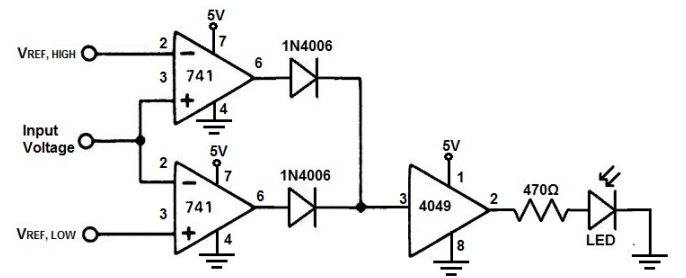


Fig. 30: Window comparator circuit diagram

A window comparator is used to determine whether an input voltage lies within a specified range, defined by an upper and a lower reference level. It is implemented by combining two op-amp comparators—one configured as a non-inverting comparator and the other as an inverting comparator.

In this design, the desired operating voltage is centered around 3.3 V, with acceptable limits set between 3.1 V and 3.5 V. The upper threshold (3.5 V) is monitored using a non-inverting comparator. In this configuration, the input voltage is applied to the non-inverting terminal, while the reference voltage (3.5 V) is applied to the inverting terminal. As a result, the comparator output goes high when the input voltage exceeds 3.5 V, effectively detecting an over-voltage condition.

Similarly, the lower threshold (3.1 V) is monitored using an inverting comparator. Here, the input voltage is applied to the inverting terminal, and the reference voltage (3.1 V) is applied to the non-inverting terminal. This configuration ensures that the output goes high when the input voltage drops below 3.1 V, indicating an under-voltage condition.

The outputs of these two comparators are combined using a diode-based OR configuration, in which each comparator output is connected through a diode to a common node. This arrangement ensures that if either comparator indicates a threshold violation, the corresponding diode conducts and pulls the combined output high. Thus, a high output is produced whenever the input voltage falls outside the defined window (i.e., either above 3.5 V or below 3.1 V), representing a “danger” condition.

To make the system more intuitive, a NOT gate is added at the final stage to invert the logic, so that the output is high only when the input voltage lies within the safe operating range (between 3.1 V and 3.5 V).

The outputs are visually indicated using LEDs: a red LED turns ON when the voltage falls outside the permissible window (danger condition), while a blue LED turns ON when the voltage lies within the safe range of 3.1 V to 3.5 V.

TABLE III: Window Comparator Truth Table

Condition on V_{in}	Danger	Safe	LED Indication
$V_{in} < 3.1 \text{ V}$	1	0	Red ON, Blue OFF
$3.1 \text{ V} \leq V_{in} \leq 3.5 \text{ V}$	0	1	Red OFF, Blue ON
$V_{in} > 3.5 \text{ V}$	1	0	Red ON, Blue OFF

In LTspice, only the window comparator stage was simulated to verify the threshold detection behavior. The final hardware implementation includes an additional inverter stage for logic inversion and LED indication of the safe and danger conditions.

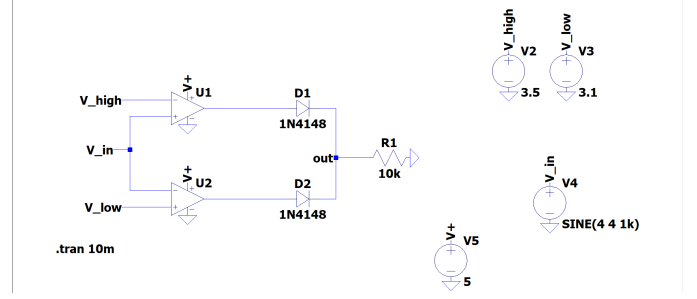


Fig. 31: LtSpice circuit for window comparator

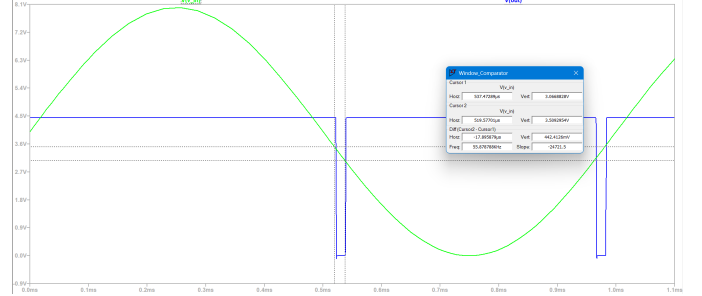
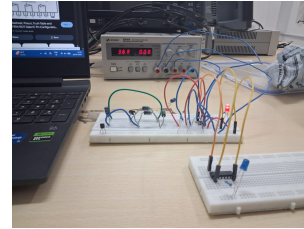
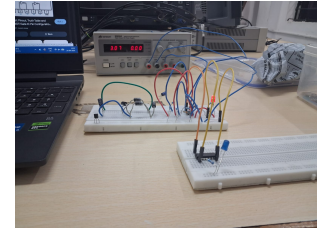


Fig. 32: LTspice simulations of the Window Comparator

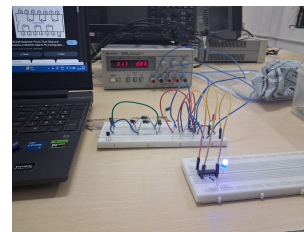
We can see from the above plot that the output goes to low in the region of 3.1 V to 3.5 V. This is the immediate output of the window comparator indicating the “danger” region.



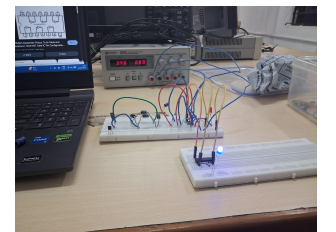
(a) V_{in} just above 3.5 V



(b) V_{in} just below 3.1 V



(c) V_{in} just above 3.1 V



(d) V_{in} just below 3.5 V

Fig. 33: Hardware Window Comparator Output at Edge cases

The hardware and simulation results match as shown above.

In the hardware implementation, the reference voltages of 3.1 V and 3.5 V are generated using a resistor divider network derived from a regulated 5 V supply provided by the Arduino. This supply is used only for establishing stable reference thresholds for the window comparator.

The reference voltages are derived using resistive voltage dividers from the Arduino's regulated 5 V output. For a voltage divider with output $V_{ref} = V_{CC} \cdot \frac{R_2}{R_1 + R_2}$, the resistor values are selected from the standard E24 series as follows:

- **Lower threshold (3.1 V):** $R_1 = 20 \text{ k}\Omega$, $R_2 = 33 \text{ k}\Omega$

$$V_{low} = 5 \times \frac{33}{20 + 33} \approx 3.11 \text{ V}$$

- **Upper threshold (3.5 V):** $R_1 = 15 \text{ k}\Omega$, $R_2 = 36 \text{ k}\Omega$

$$V_{high} = 5 \times \frac{36}{15 + 36} \approx 3.53 \text{ V}$$

Both pairs use standard resistor values and achieve the target thresholds within $\pm 1\%$, well within the tolerance of the window comparator's operating range.

It is important to note that the piezoelectric system in this project is intended for energy harvesting, where the generated voltage is inherently variable and dependent on mechanical excitation. In practical applications, piezoelectric sources are typically used to supplement existing power systems rather than operate as completely standalone replacements. Therefore, the use of a stable external reference for threshold detection does not contradict the fundamental objective of the system.

Furthermore, the resistor divider network draws negligible current (on the order of nW as verified through LTspice simulations), resulting in near-zero power consumption. As such, the reference generation stage does not impose any significant load on the system and serves purely as a low-power sensing mechanism for safe voltage operation.

Hence, the use of an external reference voltage is justified as it ensures reliable and accurate window detection while maintaining the overall efficiency and intent of the piezoelectric energy harvesting system.

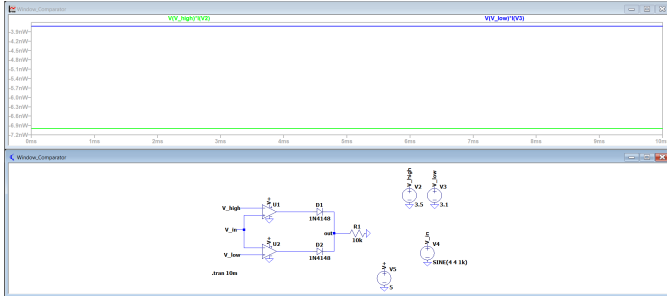


Fig. 34: Power consumed by reference voltages (in the order nanoWatts)

VIII. FULL INTEGRATED CIRCUIT

Fig. 36 shows the complete hardware implementation of the proposed system. The design follows a modular architecture, where each stage performs a specific function in the overall energy harvesting and regulation process.

The system consists of the following stages implemented in the same order:

- Piezoelectric sensors Array

- Rectifier
- Buffer
- Buck Converter with Controller
- Window Comparator

The figure below shows the complete block diagram of the implementation.

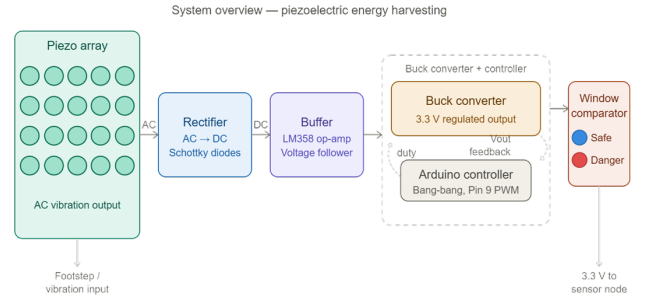


Fig. 35: Block Diagram of the Circuit

The system begins with the piezoelectric array, which converts mechanical energy into an AC electrical signal. This is followed by an AC–DC rectification stage that converts the generated signal into a usable DC voltage. Since the piezoelectric source exhibits high output impedance, an op-amp based buffer (unity-gain voltage follower) is introduced to provide impedance matching and prevent loading effects.

The buffered voltage is then fed into a buck converter stage, which steps down the varying DC input (typically in the range of 0–10 V) to a regulated 3.3 V output. This stage is controlled using a feedback-based controller implemented using Arduino microcontroller, ensuring stable output despite fluctuations in the input.

Finally, a window comparator is employed to monitor the regulated output voltage and determine whether it lies within a predefined safe operating range (3.1 V to 3.5 V). The output of this stage is used for visual indication of system status.

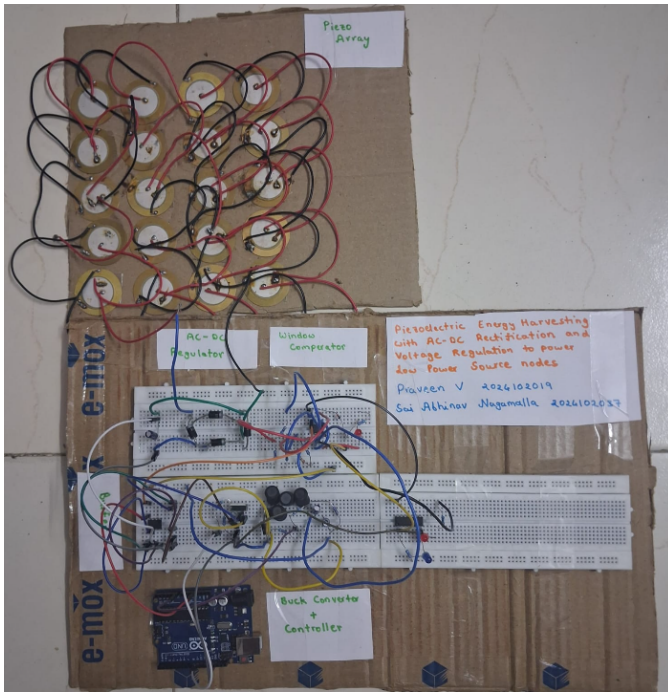


Fig. 36: Full hardware implementation of the piezoelectric energy harvesting system

Demonstration videos

The working of the full circuit is demonstrated through the videos titled *Final* which can be accessed [here](#).

IX. NOVELTY

The proposed system incorporates two key novelties that distinguish it from conventional piezoelectric energy harvesting implementations.

A. Efficient Voltage Regulation via Buck Converter

Voltage regulation is a critical requirement in any energy harvesting pipeline, as the harvested voltage must be conditioned to a stable, usable level before powering a load. Conventional regulation approaches, such as linear voltage regulator ICs (e.g., LM7805, LM317) or op-amp based regulators, operate by dissipating the excess input voltage as heat. This linear dissipation leads to significant power loss, which is particularly detrimental in a self-powered system where available energy is inherently limited.

In the proposed design, a **PWM-controlled buck converter** is employed in place of a linear regulator. The buck converter operates as a switching regulator, transferring energy in discrete pulses rather than continuously dissipating the voltage difference. This results in a theoretical switching efficiency of nearly 90%, compared to 40–60% typically achieved by linear regulators under similar conditions. Furthermore, the minimal heat dissipation characteristic of the buck converter makes it particularly well-suited for compact, self-powered embedded systems where thermal management is a constraint.

The converter is controlled by an Arduino Uno microcontroller implementing a bang-bang control algorithm. Timer1 is

configured to generate a 50 kHz PWM signal on Pin 9, with the duty cycle dynamically adjusted based on feedback from a resistive voltage divider at the output. A 16-sample ADC averaging routine is employed to suppress noise in the feedback path, and a software undervoltage lockout mechanism prevents erroneous regulation when the input falls below a minimum threshold of 1.0 V.

B. Op-Amp Based Window Comparator for Safety Indication

A second novelty of the proposed system is the integration of an op-amp based window comparator at the regulated output stage. Unlike a simple threshold comparator that detects only over-voltage or under-voltage conditions independently, a window comparator simultaneously defines both an upper and a lower voltage boundary, flagging the output as valid only when it lies within the prescribed window.

In this implementation, the window comparator continuously monitors the 3.3 V regulated output and provides real-time visual indication of the operating state:

- **Blue LED** — illuminates when the output voltage lies within the safe operating range, confirming correct regulation
- **Red LED** — illuminates when the output voltage falls outside the defined window, indicating an under-voltage or over-voltage fault condition

This safety indication mechanism adds a layer of observability to the system that is absent in most energy harvesting prototypes, enabling immediate identification of fault conditions without requiring external measurement instrumentation. It is particularly relevant in unattended sensor node deployments where real-time monitoring of power supply integrity is essential.

X. ANALYSIS AND LIMITATIONS

A. Performance Analysis

The system successfully demonstrates energy harvesting, rectification, buffering, and regulated output generation. The output voltage is maintained at approximately 3.3 V with a deviation of $\pm 1\%$, indicating satisfactory performance under dynamic input conditions. The system delivers power on the order of **hundreds of milliwatts** under peak excitation conditions, which is sufficient for powering low-power sensor nodes.

The harvested output power is inherently dependent on the amplitude and frequency of mechanical excitation applied to the piezoelectric array. Due to the intermittent and low-energy nature of such inputs (e.g., footsteps), the instantaneous power output varies significantly. As a result, the system is best suited for low-power applications with intermittent duty cycles.

Typical low-power sensor nodes require power in the range of a few milliwatts, depending on the sensing, processing, and communication requirements. This is sufficient for duty-cycled operation of ultra-low-power sensors and wireless transmission modules.

B. Sources of Error and Non-Idealities

Several non-ideal factors affect the performance of the system:

- **Component Tolerances:** Resistors used in voltage dividers and reference generation introduce errors due to tolerance (typically $\pm 1\%$ or $\pm 5\%$), affecting threshold accuracy and sensed voltage.
- **Diode Voltage Drop:** The rectifier stage introduces a forward voltage drop (approximately 0.2–0.3 V per diode), reducing the effective harvested voltage, especially at low input levels.
- **Piezoelectric Source Impedance:** The piezoelectric element behaves as a high-impedance source with capacitive characteristics. Any mismatch or loading significantly reduces the extractable power.
- **Switching Losses:** The buck converter introduces switching and conduction losses, reducing overall efficiency, particularly at low power levels.
- **Measurement Limitations:** ADC quantization and noise in the Arduino introduce small errors in voltage sensing and control decisions.

C. Design Limitations

- **Buck Controller:** Commercially available solutions often exist in the form of evaluation boards, which are optimized for performance but are typically expensive. Alternatively, a custom PCB implementation using a suitable controller IC provides a cost-effective and compact solution, allowing full customization of the control circuitry. However, designing and fabricating such hardware was beyond the scope of this course project.
- **Use of Arduino:** The Arduino-based controller is not power-efficient and requires an external supply, making it unsuitable for fully autonomous deployment. It is used here solely for validation of the control strategy.
- **Lack of Cold-Start Optimization:** The current implementation does not include a dedicated cold-start mechanism, which is critical for real-world energy harvesting systems.
- **Diode-Based Rectification:** The use of standard diodes limits efficiency at low voltages due to forward voltage drops.
- **Absence of Energy Storage:** The system does not incorporate a storage element (e.g., supercapacitor or rechargeable battery), limiting its ability to supply continuous power during low or no excitation.

D. Actual Deployment Pitfalls

While the control algorithm is validated using a microcontroller, implementing the same control strategy in a dedicated hardware IC or custom PCB presents several challenges:

- Designing stable feedback compensation networks for analog controllers requires careful tuning and analysis of system dynamics.
- Integrating control logic (e.g., hysteretic or PID-based control) into hardware requires additional circuitry such

as comparators, ramp generators, and compensation networks.

- Ensuring low quiescent current operation is critical in energy harvesting systems, which is non-trivial in discrete implementations.
- Commercial evaluation boards that provide such functionality are often expensive, while custom PCB design requires expertise and fabrication resources beyond the scope of this project.

E. Suggestions for Improvement

- Replace diode rectification with synchronous rectification to improve efficiency.
- A buck–boost converter can be employed instead of a simple buck converter to regulate both higher and lower input voltages to a stable 3.3 V output. However, implementing a stable control scheme for a buck–boost topology is more complex due to its non-linear dynamics and wider operating range. Hence, a buck converter was chosen in this work for simplicity and ease of implementation within the project scope.
- Incorporate energy storage elements (supercapacitors or thin-film batteries) to enable continuous operation.
- Implement maximum power point tracking (MPPT) techniques to maximize energy extraction from the piezoelectric source.
- Optimize the mechanical coupling of the piezoelectric element to improve energy conversion efficiency.

XI. CHALLENGES FACED

The development of the system involved several practical and conceptual challenges:

- **Understanding the buck converter and controller:** Gaining a clear understanding of the working principles of the buck converter and its control mechanism required significant effort, particularly in relating theoretical concepts to practical implementation.
- **Soldering complexity:** The hardware implementation required soldering a 4×5 array of piezoelectric sensors, which was time-consuming and required precision to ensure proper connections and reliable performance since the piezo sensors were too sensitive to high temperatures used for soldering.
- **Low and intermittent power generation:** The electrical output from the piezoelectric sensors is inherently low and highly dependent on mechanical excitation. This made it challenging to obtain a stable and sufficient input for further processing stages.
- **Voltage loss in rectification:** The AC–DC conversion stage introduces voltage drops due to the forward voltage of diodes, significantly reducing the effective usable output, especially at low input levels.
- **Time constraints:** The required components were requested at the outset of the project. However, delays in their procurement significantly reduced the available time. This directly limited the scope for iterative design,

thorough testing, and optimization. Consequently, the implementation was constrained to a design with simple novelty, while more advanced features and refinements could not be fully realized within the given timeframe.

- **Limited reference material:** The availability of relevant and directly applicable literature for piezoelectric energy harvesting systems with integrated regulation and control was limited. This necessitated independent design decisions and increased the effort required for validation and troubleshooting.

XII. FURTHER/ONGOING RESEARCH

Recent research in piezoelectric energy harvesting is rapidly advancing, particularly in countries like Japan and the United States, with a focus on improving efficiency, scalability, and real-world applications.

One notable development is the use of piezoelectric materials in public infrastructure. Japan has experimented with embedding piezoelectric tiles in high-footfall areas such as railway stations, where mechanical energy from footsteps is converted into electrical energy. This demonstrates the potential of integrating energy harvesting into smart city environments (Watch demo).

Another important area of research is vibration-based energy harvesting. Modern studies focus on extracting energy from low-frequency ambient vibrations (0–100 Hz), which are commonly found in real-world environments such as vehicles, buildings, and industrial systems. These systems are particularly useful for powering small, wireless sensors (View paper).

Overall, current project in hand aims to overcome key challenges such as low power output and efficiency limitations, with the ultimate goal of enabling battery-free IoT devices and sustainable energy solutions.

XIII. CONCLUSION

In this project, a piezoelectric energy harvesting system was successfully designed and implemented to generate electrical power from ambient mechanical vibrations. By utilizing an array of piezoelectric sensors, the inherently low and irregular output was improved in terms of both voltage and current. The generated AC signal was effectively converted into a usable DC output through a full-wave rectifier, followed by filtering to reduce ripple.

A buck converter was employed to regulate the output voltage to a stable 3.3 V, ensuring a supply suitable for low-power electronic applications. The control algorithm, implemented on an Arduino Uno, employs a hysteretic bang-bang strategy with ADC-based voltage feedback, achieving a regulation deviation of $< 1\%$. A window comparator was integrated at the output stage to provide real-time visual safety indication, illuminating a blue LED when the output lies within the safe operating range and a red LED upon detection of an under-voltage or over-voltage fault condition.

Although the harvested power is relatively small, the results demonstrate the feasibility of piezoelectric energy harvesting

as a sustainable alternative to conventional power sources for low-power devices. The system highlights the potential for integration into self-powered IoT applications, wearable electronics, and environments where battery replacement is impractical.

XIV. ACKNOWLEDGMENT

We would like to express my sincere gratitude to our instructors, Dr. Priyesh Shukla and Dr. Arti for proposing this project.

We also extend our thanks to our teaching assistants, Pranav and Ritama, for their support and assistance. Special thanks to Akshat for providing timely resources that greatly helped in completing this project.

Working on this project has been a valuable learning experience.

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